Kairo Master Plan v0.5

> Single source of truth untuk seluruh perencanaan Kairo Runtime, Kairo Simulator, dan Kairo Board.

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1. Vision

Kairo adalah platform handheld device yang berfokus pada:

- Hardware hacking
- Wireless experimentation
- Plugin ecosystem
- Developer experience
- Extensible runtime architecture
- Simulator-first development

Targetnya bukan sekadar membuat perangkat mirip Flipper Zero.

Tujuan jangka panjang adalah membangun sebuah platform yang memiliki:

- Runtime yang portable
- Plugin system yang kuat
- Hardware abstraction yang baik
- Simulator untuk development cepat
- Dukungan multi-device di masa depan

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2. Official Naming

Runtime

Kairo Runtime

Core operating/runtime layer yang berjalan pada seluruh perangkat Kairo.

SDK

Kairo SDK

Framework untuk mengembangkan:

- Apps
- Plugins
- Services

Simulator

Kairo Simulator

Environment development dan debugging berbasis web.

Hardware

Kairo Board

Nama resmi hardware Kairo.

Contoh:

- Kairo Board DevKit S3
- Kairo Board V1
- Kairo Board V2

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3. Core Principles

Simulator First

Semua fitur sebisa mungkin dapat dikembangkan dan diuji tanpa hardware.

Observability First

Debugging adalah prioritas.

Runtime harus mudah diinspeksi.

Plugin First

Fitur tambahan harus dapat dikembangkan sebagai plugin.

Hardware Agnostic

Core Runtime tidak boleh mengetahui hardware tertentu.

Capability Driven

Aplikasi tidak memeriksa jenis hardware.

Aplikasi memeriksa capability.

Contoh:

```
if (capabilities.has("wifi"))
```

Bukan:

```
if (isEsp32())
```

---

4. Repository Structure

```
kairo/

firmware/
├─ core/
├─ platforms/
├─ boards/
├─ targets/
└─ tools/

packages/
└─ simulator-web/

docs/

package.json
bun.lock
```

---

5. Architecture Hierarchy

```
Core
 ↓
Platform
 ↓
Board
 ↓
Target
```

---

6. Core Runtime

Core Runtime adalah bagian terpenting dari sistem.

Core tidak mengetahui:

- ESP32
- Display tertentu
- WiFi tertentu
- Board tertentu

Core hanya mengenal interface.

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7. Core Services

Logger Service

Service pertama yang harus tersedia.

Semua subsystem wajib menggunakan logger.

Tidak boleh menggunakan:

```
printf()
```

secara langsung.

Log Levels

- TRACE
- DEBUG
- INFO
- WARN
- ERROR
- FATAL

Log Entry

Contoh:

```
[12:00:01] [INFO ] [Runtime] Booting

[12:00:02] [INFO ] [Logger] Initialized

[12:00:03] [ERROR] [Wifi] Connection Failed
```

Structured Logging

Contoh:

```json
{
  "level": "ERROR",
  "component": "WifiService",
  "message": "Connection Failed",
  "ssid": "Office"
}
```

Log Sinks

- Console Sink
- Memory Sink
- File Sink
- Remote Sink (future)

Event Bus

Backbone komunikasi internal.

Example Events

- SystemBoot
- SystemReady
- ServiceStarted
- ServiceStopped
- PluginLoaded
- PluginUnloaded
- NotificationCreated
- BatteryChanged
- NetworkConnected
- NetworkDisconnected

Service Container

Dependency Injection Container.

Responsibilities:

- Register Service
- Resolve Service
- Singleton Management
- Lifecycle Management

Service Manager

Mengelola lifecycle service.

States:

- Created
- Starting
- Running
- Stopping
- Stopped
- Failed

Plugin Manager

Mengelola:

- Loading
- Unloading
- Lifecycle
- Permissions
- Registration

Notification Manager

Mengelola notifikasi sistem.

Network Manager

Abstraksi seluruh konektivitas jaringan.

Aplikasi tidak berbicara langsung ke driver.

Flow:

```
 App
  ↓
Network Manager
  ↓
Network Adapter
  ↓
Driver
```

Display Manager

Abstraksi display.

Aplikasi tidak berbicara langsung ke display driver.

Storage Manager

Abstraksi storage.

Mendukung:

- Internal Flash
- SD Card
- Future Storage Providers

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8. System Introspection

SystemInfo

Memberikan informasi runtime.

Contoh:

- CPU
- RAM
- PSRAM
- Flash
- Build Version
- Firmware Version
- Board Version

Hardware Registry

Mendaftarkan hardware yang tersedia.

Contoh:

- Display
- WiFi
- Bluetooth

- Battery
- Speaker
- Microphone
- NFC
- RFID
- Infrared
- SubGHz
- GPS
- SD Card

Capability Registry

Mendaftarkan kemampuan yang tersedia.

Contoh:

- Networking
- Notifications
- OTA
- Plugin Runtime
- Script Runtime
- Background Services

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9. Platform Layer

Platform adalah implementasi runtime environment.

Simulator Platform

Digunakan untuk simulator.

Menyediakan:

- Mock Wifi
- Mock Bluetooth
- Mock Display
- Mock Storage
- Mock Battery

ESP32 Platform

Digunakan untuk perangkat nyata.

Menyediakan:

- ESP Wifi
- ESP Bluetooth
- ESP Display
- ESP Storage
- ESP Battery

---

10. Board Layer

Board mendefinisikan hardware.

Simulator Board

Virtual hardware.

DevKit S3 Board

Development board.

Current hardware:

ESP32-S3-WROOM-1-N8R8

Purpose:

- Runtime Development
- Driver Development
- Plugin Testing
- Integration Testing

Kairo Board V1

Production hardware target.

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11. Target Layer

Target adalah entry point firmware.

Simulator Target

```
 Target
  simulator

Board
  simulator

Platform
  simulator
```

DevKit Target

```
 Target
  devkit-s3

Board
  devkit-s3

Platform
  esp32
```

Kairo Board V1 Target

```
 Target
  kairo-board-v1

Board
  kairo-board-v1

Platform
  esp32
```

---

12. Boot Flow

```
 main()

  ↓

Create Runtime

  ↓

Load Platform

  ↓

Load Board

  ↓

Register Services

  ↓

Initialize Core

  ↓

Start Runtime

  ↓

Launch Home Screen
```

---

13. UI Architecture

```
 Plugin
  ↓
Screen
  ↓
UI Runtime
  ↓
Display Manager
  ↓
Display Adapter
  ↓
Display Driver
  ↓
Hardware
```

---

14. UI Runtime

Standar UI resmi Kairo.

Plugin tidak boleh menggambar pixel secara langsung.

Standard Components

- Screen
- Label
- Button
- List
- Menu
- Dialog
- Notification
- StatusBar
- ProgressBar
- Icon

---

15. Screen System

Plugin membuat screen.

Plugin tidak menggambar display.

Flow:

```
 Plugin
  ↓
Screen
  ↓
UI Runtime
  ↓
Display Manager
  ↓
Display Driver
```

---

16. Status Bar

Core Status Items:

- WiFi
- Bluetooth
- Battery
- Clock

Plugin Status Items:

Plugin dapat menambahkan icon ke status bar.

Contoh:

- VPN
- Script Running
- Background Service

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17. Background Services

Foreground:

Hanya satu aplikasi aktif.

Background:

Banyak service dapat berjalan bersamaan.

Contoh:

- Clock Service
- Notification Service
- Wifi Service
- Bluetooth Service

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18. Simulator

Simulator bukan emulator hardware.

Simulator menjalankan:

- Kairo Runtime
- Plugin Runtime
- UI Runtime
- Mock Drivers

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19. Kairo Simulator

Location:

```
packages/simulator-web
```

Technology:

- Bun
- TypeScript
- React
- Vite
- WebSocket

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20. Simulator Panels

Display

Render virtual display.

Logs

Realtime logs.

Filter:

- TRACE
- DEBUG
- INFO
- WARN
- ERROR
- FATAL

Events

Realtime event stream.

Services

Realtime service lifecycle monitor.

Hardware Registry

Hardware visibility.

Capability Registry

Capability visibility.

Controls

Actions:

- Boot
- Shutdown
- Restart
- Factory Reset
- Install Plugin
- Remove Plugin
- Toggle WiFi
- Toggle Bluetooth
- Inject Event
- Change Battery Level

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21. Kairo Board V1

Goal

Minimal capability target harus setara dengan Flipper Zero.

Dengan runtime dan plugin ecosystem yang lebih modern.

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22. Kairo Board V1 Hardware Targets

Processing

Candidate:

- ESP32-S3
- ESP32-P4 + Companion Radio MCU

Decision TBD.

Memory

Minimum Target:

Flash:

```
32 MB
```

PSRAM:

```
8 MB
```

Storage:

```
MicroSD
```

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23. Connectivity

WiFi

Required

Use Cases:

- OTA
- Plugin Download
- Cloud Features
- Remote Control

Bluetooth

Required

Use Cases:

- Mobile Companion
- BLE Tools
- Device Pairing

---

24. Radio Features

SubGHz

Required

Target Bands:

- 315 MHz
- 433 MHz
- 868 MHz
- 915 MHz

NFC

Required

Target:

13.56 MHz

RFID

Required

Target:

125 kHz

Infrared

Required

Features:

- RX
- TX

---

25. Audio

Speaker

Required

Microphone

Required

26. Storage

MicroSD

Required

Use Cases:

- Plugins
- Scripts
- Logs
- Media
- Backups

27. Power

Required:

- Rechargeable Battery
- USB-C Charging
- Battery Monitoring
- Battery Protection

28. Input

Required:

- Navigation Buttons
- Action Button
- Back Button
- Power Button

Layout TBD.

29. Display

Target:

Color IPS Display

Requirements:

- Good readability
- Low power consumption
- Fast refresh rate

30. Expansion

Reserved expansion capability.

Potential future modules:

- GPS
- LoRa
- External Radios
- Custom Add-ons

31. Mandatory Capabilities

Kairo Board V1 wajib memiliki:

- WiFi
- Bluetooth
- NFC
- RFID
- Infrared
- SubGHz
- Speaker
- Microphone
- MicroSD
- USB-C
- Battery Monitoring
- Plugin Runtime
- OTA Updates
- Background Services

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32. Development Roadmap

Milestone 1

Core Foundation

- Logger
- Event Bus
- Service Container
- Boot Sequence

Milestone 2

Observability

- Log Viewer
- Event Viewer
- Service Inspector

Milestone 3

Simulator

- Simulator Platform
- Simulator Board
- Simulator Web

Milestone 4

Plugin Runtime

- Plugin Manager
- Plugin Lifecycle
- Plugin API

Milestone 5

UI Runtime

- Status Bar
- Screen System
- Home Screen

Milestone 6

ESP32 Dev Hardware

Target:

DevKit S3

Board:

ESP32-S3-WROOM-1-N8R8

Milestone 7

Kairo Board V1

- Hardware Design
- PCB Design
- Prototype
- Validation

Current Development Hardware

Board:

Kairo Board DevKit S3

MCU:

ESP32-S3-WROOM-1-N8R8

Flash:

8 MB

PSRAM:

8 MB

Purpose:

- Runtime Development
- Driver Testing
- Integration Testing
- Simulator Validation

Before Kairo Board V1 exists.